The State of AI

D-Lab AI Pulse, Fall 2026: Session 1

Materials: `https://github.com/dlab-berkeley/AI_Pulse`
Workshop suggestions: `https://forms.gle/x8KfejgCViE7t6iQA`

1 September 2026

Today's Roadmap

- AI Pulse is D-Lab's bi-weekly series on AI tools for academia, no experience required!
- Part 1, where things stand, what changed, what exists now, and what it costs
- Part 2, what agents can do, and are these tools actually working?
- Part 3, how your role changes, what to hand over, and what that leaves you doing

This Semester

Eight sessions, 1 September to 8 December. Every other Tuesday.

- Workshops alternate: "wide-view" sessions, and hands-on sessions with demos
- Office hours on the weeks in between, same day and time, on Zoom
- Office hours are for your actual problem: getting a tool working, or working out which one fits what you are trying to do
- Coming up: The Major AI Ecosystems (15 Sept) and AI in the News (29 Sept)
- Materials: `https://github.com/dlab-berkeley/AI_Pulse` · D-Lab: `https://dlab.berkeley.edu`

Part 1

Where Things Stand

How We Got Here

- 2017: *Attention Is All You Need*, the architecture behind all of it
- 2020: AlphaFold 2 solves protein folding, earning its authors the 2024 Nobel Prize in Chemistry
- 2022: ChatGPT opens to the public, five years after the paper
- 2025: Claude Code. AI goes from chats to agents
- May 2026: the first AI-produced proof of a major open problem. An 80-year-old Erdős conjecture, reasoned through rather than searched for

Timothy Gowers, Fields Medallist: *"if a human had written the paper and submitted it to the Annals of Mathematics . . . I would have recommended acceptance without any hesitation."*

From Chat to Agents

- 2023, chats
  - A powered-up search engine. You asked; you copied the answer out by hand
- 2024, files
  - It could read your documents, and hand one back finished
- 2025, tasks
  - You describe the job, walk away, come back to the result
- 2026, agents
  - A whole team. Your job becomes that of a manager

Each step changed what you can ask for, not just how much you can hand over.

Fifteen Claudes at Once

Boris Cherny, who built Claude Code, on a normal working day:

- Fifteen running at once: five in terminals, five to ten in the browser, one on his phone
- Each on its own task, in its own copy of the project, so nothing collides
- They test their own work before handing it back; he reviews and decides what is kept
- Ten to thirty finished pieces of work merged per day

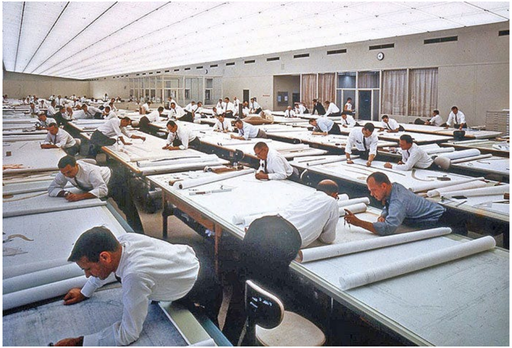
At Anthropic, 80% of the code merged into production in May 2026 was written by Claude.

Meanwhile, Everybody Else

- One billion people used Google's Gemini app last month, the fastest growth of any product Google has shipped. ChatGPT is in the same range
- Half of US employees used AI at work this year, up from 46% a year ago. More than a quarter use it every week
- At ICLR 2026, the largest conference in machine learning, 21% of the 70,000 peer reviews were fully AI-written, and over half showed some AI
- 95% of UK undergraduates use AI, 94% of them on assessed work
- Job ads asking for AI skills grew 69% in a year, while hiring overall grew 9%

Is it a way of standing out, or is it necessary not to lag behind?

How Is Work Changing?



- Did engineers disappear when CAD was released?
- Could engineers keep their jobs without mastering the new tech?
- How long did it take for CAD to become necessary, and how long will it take for AI?

What Is Actually Out There

- General assistants: ChatGPT, Claude, Gemini, Grok
    - They all do the same things now. Choose on fit, not on capability
- Coding platforms: Cursor, Claude Code, Devin, Copilot
    - Built for code, but increasingly just run tasks. A platform, not a chatbot
- Research tools: NotebookLM, Elicit, Consensus
    - Finding, reading and synthesizing literature, with the sources attached
- Generation: Midjourney, Veo, ElevenLabs, Lovable
    - Images, video, voice, websites. One job each, done well
- Productivity: Granola, Otter, Notion
    - Ready-made. They take the grunt work without you designing anything

What It Costs

- Free with your Berkeley account: Gemini, NotebookLM and Gemini CLI*
  - Under the campus agreement your data does not train the model, and it reaches your Drive, Docs, Gmail and Calendar
- OpenAI: Free · Go $8 · Plus $20 · Pro $100 or $200 a month
  - Free is chat; Go adds messages, uploads and voice; Plus adds deep research and projects; Pro adds the reasoning models and research previews
- Anthropic: Free · Pro $20 · Max $100 or $200 a month
  - Free is chat; Pro adds Claude Code, Cowork and the Microsoft 365 integration; Max adds usage limits and priority access

Next session: OpenAI vs Anthropic in depth, with demos.

* *Evolving situation.*

What a Token Costs

- A token is the fundamental unit of information for AI: roughly three quarters of a word
- You pay for what you send in, and again for what comes back
- Output costs roughly five times input, in every model, from both vendors

Anthropic			OpenAI		
Model	In	Out	Model	In	Out
Fable 5	$10	$50	GPT-5.6 Sol	$5	$30
Opus 5	$5	$25	GPT-5.6 Terra	$2	$12
Sonnet 5	$2	$10	GPT-5.6 Luna	$0.20	$1.20
Haiku 4.5	$1	$5			

US dollars per million tokens. September 2026.

Older Models Are Good Enough

- On LMArena, where people vote blind on which of two answers they prefer, the top five models sit within 10 points of each other
- Second place goes to Claude Opus 4.6, a model two generations old, two points behind the leader and at half the price
- Gemini 3.7 Flash sits ninth, 17 points off the top, at a thirteenth of the leader's input price
- What decides is interface, price, privacy rules, and what your institution already pays for
- We have reached a stage in which the best option is not the best model

Part 2

What Agents Do, and Whether It Works

What Makes It an Agent

- A chat produces text. You are still the one who acts on it
- An agent can open, read, write, run and check things itself
- The model underneath is the same. What is new is access and permission, not intelligence
- It started with code because code checks itself: it runs or it does not, and the tests pass or they fail
- So the model can try, fail and fix before a human sees anything, and keep going until the task is done or it needs you

What an Agent Can Reach

- Your files: read them, edit them, write new ones
- Your calendar, mail and notes, through MCP, the common standard that lets tools connect to each other
- Code: write it, run it, read the error, fix it, run it again
- The internet: search it, open pages, and use what it finds
- Your team's tools: GitHub, Slack, Jira and the like, through that same standard
- Time and place: start a task from your phone, or schedule it to run every day

Beyond the Main AI Apps

- Context is the constraint for repeated, independent tasks: an agent carries one conversation, and a thousand items need a thousand clean starts
- So for the same operation across thousands of items, you call the model directly through its API instead
- A gateway such as OpenRouter or LiteLLM sends that call to any model, and lets you switch models without rewriting anything
- Four thousand PDFs, one extraction schema, one spreadsheet of results
- You can also adapt an open model such as Llama to one narrow task: sensitive data stays on your machine, cost per document falls, and the model can be specialized

Built for Science

- Elicit: works across the literature, searching, screening and extracting at systematic-review scale
- SciSpace: sits inside one paper with you. Chat with the PDF, get a figure or a derivation explained, pull the numbers out
- AI Co-Scientist: proposes hypotheses and designs the experiments to test them. Published in *Nature* in May, validated at the bench

Two sessions on this later in the semester: AI in Science, Workflows, and AI in Science, Tools.

Developing: on 27 August Anthropic released the Model Hardware Standard, which is MCP for physical instruments. Robotic arms, lab equipment, manufacturing machines.

Nobody Stays in Front

- The lead changes every few months. Whoever tops the table today has usually been passed by the next session of this workshop
- In July, Moonshot AI released Kimi K3: 2.8 trillion parameters, the largest openly released model to date. It placed first on one major coding leaderboard and third on another
- Chinese labs are now at the frontier, and the open-weight ones, DeepSeek, Qwen and GLM among them, remain far cheaper than the American models
- Open weights matter more than headline price: a model you run yourself costs nothing per use, and sends nothing anywhere
- So weigh the cost of switching against the gain from always being on the best

Changing the Question

- Everything so far has been about what these tools can do, and most of it comes from the people who make them
- Very little of it comes from a study designed to find out whether the tools actually help
- So the question changes: when somebody measures carefully, what do they find?
- Three studies: a state labor department, a hospital, and a peer review process

What Happens When Somebody Measures

- When you file for unemployment insurance, an adjudicator reviews your claim and writes back asking for whatever is missing. It is slow, and it is repetitive
- Colorado's labor department and Stanford fine-tuned a model on the department's own records to draft those follow-up questions
- They ran a randomized trial: 788 drafts, eight adjudicators, blind quality review against two control groups
- Time saved per case: four seconds. Statistically indistinguishable from zero
- Quality, against the adjudicator working alone: no significant difference. They published the result and did not deploy the system

The Failure You Cannot See

- A model does not know what it does not know
- In a 1,100-case medical safety study, over 80% of severe errors were things the model left out, not things it invented
- A UK government review of public consultation responses found the same shape: human reviewers used the "Other" category seventeen times more often than the model did
- It does not say that it is unsure. It files the case under the nearest heading it already has, and nothing in the output shows that anything is missing

Accuracy Is Not Safety

- Two studies of AI in clinical medicine, months apart, reached opposite headlines. Both are correct
- Harvard and Stanford: 76 emergency cases, one model. It matched or exceeded expert physicians on diagnosis
- NOHARM: 1,100 consultations, 24 systems. Potential for severe harm in up to a quarter of cases
- One measured whether it gets the answer right. The other measured what happens when it does not
- The average and the tail are different questions, and only one of them made the headlines

Part 3

How Your Role Changes

Replacing Workers, or Replacing Work?

- The question people ask is whether these tools replace people
- A more immediate question is: which parts of a job they take over, and which parts they leave alone
- Every technology that automated a task changed what the job was long before it changed how many jobs there were
- So the question is not whether your field survives this, but which hours of your week are the ones under discussion

The Centaur

- In May 1997, Deep Blue beat Garry Kasparov. Chess was widely treated as settled
- Kasparov's response was to invent a format where humans and machines play together. Eight years later a tournament ran it open to anyone, with any machine
- It was won by two American amateurs running three ordinary PCs
- They were not better at chess. They were better at running the machines: knowing which positions to push on, and when to disagree
- Kasparov: "*weak human + machine + better process was superior to a strong computer alone*", and also to a strong human with a worse process

The Same Pattern, in Mathematics

- The Erdős problems are a public list of open problems in mathematics, kept on a website anyone can post to
- Professors and hobbyists comment side by side. Terence Tao was among the most active
- Before AI, the problems closed at about five a month
- 18 have now been fully solved by AI, and 36 more with AI assistance
- That includes one by Liam Price, who does not have a degree
- Knowing how to guide AI was enough

What That Makes Your Job

- You are not being asked to write less. You are being asked to specify, review and decide more
- Describing a task precisely is now a core skill. “Analyze this” is not an instruction
- The tools that coordinate work matter more, not less: version control, GitHub, Slack, Monday
- You no longer have to memorize the syntax. You do have to be able to tell when the answer is wrong
- Hand over the work you are time-constrained on, not the work you are cognitively constrained on
    - Ask it to check the grammar across a long document. Do not ask it to write the page for you

Food for Thought

- How do I break my work into smaller steps, and what does “correct” mean for each step?
- Centaur chess is over: the engines got strong enough that the human stopped adding anything. Which parts of your work are like chess?
- What in your work is the thinking, and what is the drafting?
- Is using this a way of standing out, or is it keeping up?
- If a model wrote the first draft, whose work is it?

Where to Go From Here

- Slides and materials: `https://github.com/dlab-berkeley/AI_Pulse`
- Workshop suggestions: `https://forms.gle/x8KfejgCViE7t6iQA`
- What Berkeley already gives you: `https://life.berkeley.edu/ai-tools-for-berkeley-students`
- D-Lab: `https://dlab.berkeley.edu`

Next: The Major AI Ecosystems, 15 September. Office hours next week, same day and time, on Zoom.

References

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- Gemini's billion users: Google, 11 August 2026. Workplace use: Gallup, Q1 2026. Student use: HEPI and Kortext, *Student Generative AI Survey 2026*, 1,054 UK undergraduates. Job ads: PwC *Global AI Jobs Barometer 2026*
- AI-written peer reviews at ICLR 2026: Pangram, 18 November 2025, over 70,000 reviews. Note that Pangram sells AI detection
- Berkeley's tools and data protections: Berkeley IT and the EVCP office, 2025; `life.berkeley.edu/ai-tools-for-berkeley-students`
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